Dallas International School

Test 2

Topics 4.2-4.4

12th IB - Math AA - SL & HL

Name : . . . . . .

EXERCISE 1 [Maximum mark : 5]

The following table shows the marks scored by seven students on two different mathematics tests.

Test 1 (x)	14	20	24	28	33	35	42
Test 2 (y)	18	25	24	30	34	33	39

Let L_1 be the regression line of x on y. The equation of the line L_1 can be written in the form $x = ay + b$.

(a) Find the value of a and the value of b. [2]

(b) Let L_2 be the regression line of y on x. The lines L_1 and L_2 pass through the same point with coordinates (p, q).

Find the value of p and the value of q. [3]

EXERCISE 2 [Maximum mark : 8]

The following table shows the systolic blood pressures, p mmHg, and the ages, t years, of 6 male patients at a medical clinic.

Patient	P1	P2	P3	P4	P5	P6
t (years)	40	72	35	47	21	61
p (mmHg)	105	145	100	130	95	132

(a.i) Determine the value of Pearson's product-moment correlation coefficient, r, for these data. [2]

(a.ii) Interpret, in context, the value of r found in part (a) (i). [1]

The relationship between t and p can be modelled by the regression line of p on t with equation $p = at + b$.

(b) Find the equation of the regression line of p on t. [2]

A 50-year-old male patient enters the medical clinic for his appointment.

(c) Use the regression equation from part (b) to predict this patient's systolic blood pressure. [2]

(d) A 16-year-old male patient enters the medical clinic for his appointment.

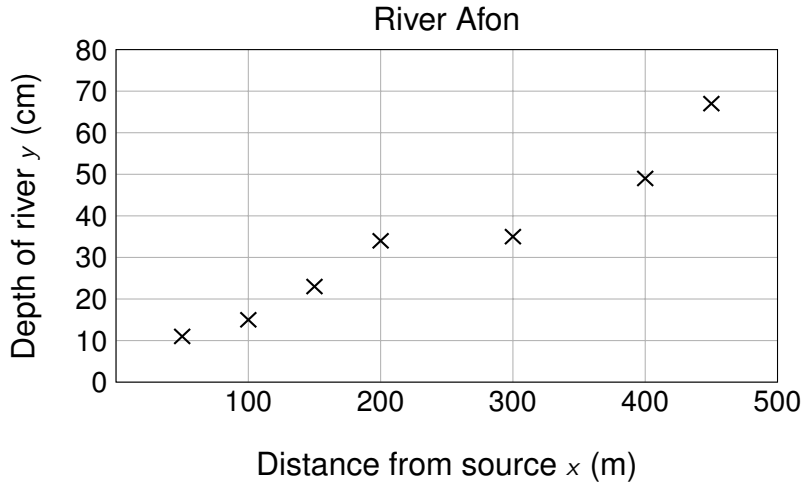
Explain why the regression equation from part (b) should not be used to predict this patient's systolic blood pressure. [1]

EXERCISE 3 [Maximum mark : 7]

Kiran and Logan collect the following data about the river Afon, where x is the distance in metres from the source and y is the depth in centimetres.

Distance from the source x (m)	50	100	150	200	300	400	450
Depth y (cm)	11	15	23	34	35	49	67

This data is represented in the following scatter diagram.



Kiran knows that the depth of the river is 0 cm at the source.

Kiran calculates $\bar{x}$ and $\bar{y}$ for the seven points given in the table, and draws a line on the scatter diagram through the mean point $(\bar{x}, \bar{y})$ and the point $(0, 0)$.

(a) Find

(a.i) the value of $\bar{x}$ and the value of $\bar{y}$; [1]

(a.ii) the equation of Kiran's line. [2]

For the seven points given in the table Logan finds the regression line of y on x with equation $y = ax + b$, where $a, b \in \mathbb{R}$.

(b) Determine the value of a and the value of b. [2]

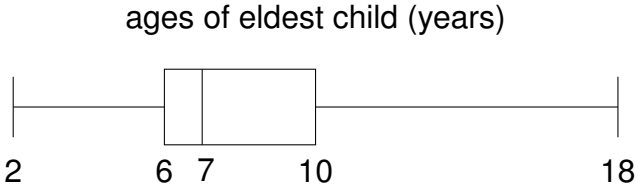
(c) By using the equation of Logan's regression line, estimate the depth of the river 350 m from its source. [2]

EXERCISE 4 [Maximum mark : 7]

A survey at a swimming pool is given to one adult in each family. The age of the adult, a years old, and of their eldest child, c years old, are recorded.

The ages of the eldest child are summarized in the following box and whisker diagram.

diagram not to scale



Note : a data value is considered an outlier if it is greater than $Q_3+1.5\times IQR$ or less than $Q_1-1.5\times IQR$.

(a) Find the largest value of c that would not be considered an outlier. [3]

The regression line of a on c is $a = \frac{7}{4}c + 20$. The regression line of c on a is $c = \frac{1}{2}a - 9$.

(b.i) One of the adults surveyed is 42 years old. Estimate the age of their eldest child. [2]

(b.ii) Find the mean age of all the adults surveyed. [BONUS : 2]